Assignment

Answer the following Questions.

1. Why loops are necessary in Programming Language?
2. What is the purpose of pass keyword?
3. What is the difference between break and continue?
4. What is the difference between while loop and for loop?

1. Why are loops necessary in Programming Language?

Ans: Loops are necessary because they allow a block of code to be executed repeatedly without writing the same code multiple times. They help reduce code redundancy, save time, and make programs more efficient.

Example:

```
for i in range(5):
    print("Hello")
```

This prints "Hello" 5 times without writing the `print()` statement repeatedly.

2. What is the purpose of the `pass` keyword?

Ans: The `pass` keyword is a null statement in Python. It is used as a placeholder when a statement is syntactically required but no action needs to be performed.

Example:

```
for i in range(5):
    if i == 3:
        pass
    print(i)
```

It allows the program to run without errors even when a code block is empty.

3. What is the difference between `break` and `continue`?

Ans:

break	continue
Terminates the loop completely.	Skips the current iteration and moves to the next iteration.
Control exits the loop.	Control remains inside the loop.

Example of `break`:

```
for i in range(5):
    if i == 3:
        break
    print(i)
```

Output:

```
0
1
2
```

Example of `continue`:

```
for i in range(5):
    if i == 3:
        continue
    print(i)
```

Output:

```
0
1
2
4
```

5. What is the difference between `while` loop and `for` loop?

Ans:

while loop	for loop
Executes as long as a condition is true.	Executes a fixed number of times or over a sequence.
Used when the number of iterations is unknown.	Used when the number of iterations is known.
Condition is checked before each iteration.	Iterates through a sequence or range automatically.

Example of `while` loop:

```
count = 1
while count <= 5:
    print(count)
    count += 1
```

Example of `for` loop:

```
for i in range(1, 6):
    print(i)
```

Both loops produce the same output, but `for` loops are generally preferred when the number of iterations is known in advance.